

Rank scientific idea suggestions generated by an AI paper-writing assistant

Your goal in this task is to rank idea suggestions written by an AI assistant. The AI assistant helps its users write paper abstracts by writing sentences with proposals for new ideas or questions to consider.

You are first given:

1. A context which describes relevant background in a specific area of interest. e.g., *background on efficiency issues in large language model training.*
2. A seed term that should be a focus of the generated scientific idea. see example below.

Then, you are given a list of 5 candidate suggestions — each generated by the AI assistant using different approaches. Each candidate is written in the form of a sentence from a scientific paper abstract.

Your goal is to score the candidate suggestions with a helpful/unhelpful answer. Your scoring decision should consider the overall quality of each candidate, using considerations that include:

1. Is the candidate relevant to the context + seed term?
2. Does the candidate copy too much from the context, or is it sufficiently novel/different from the context?
3. Does the candidate's suggestion generally make sense to you scientifically?
4. Is the language sufficiently clear and coherent for you to understand the suggestion?

The AI scientific assistant is a work in progress – don't expect it to generate highly-detailed ideas or breakthrough discoveries! Use the three guideline metrics above to come up with your overall rating.

Read the following example that illustrates the task:

Context: *the task of converting a natural language question into an executable sql query , known as text - to - sql , is an important branch of semantic parsing . the state - of - the - art graph - based encoder has been successfully used in this task but does not model the question syntax well .*

Seed term: *diverse relational edge embedding*

System Outputs:

- *We propose a novel technique , Diverse Relational Edge Embedding (DREE) , to address this problem.*
- *We propose a novel graph-based encoder that uses a diverse relational edge embedding to model the question syntax.*
- *Diverse relational edge embedding is important for text-to-SQL parsing because it can help the parser to better understand the question syntax.*
- *In this paper, we propose a novel graph - based encoder for diverse relational edge embedding.*
- *In this paper, we propose a novel graph - based encoder for text - to - sql (ttsql) that incorporates the question syntax into diverse relational edge embedding.*

Ranking:

- unhelpful
- helpful
- helpful
- unhelpful
- helpful

Figure 6: Human evaluation instructions

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Does the candidate's suggestion generally make sense to you scientifically?

Is the language sufficiently clear and coherent for you to understand the suggestion?

Context: while pretrained language models achieve excellent performance on natural language understanding benchmarks , they tend to rely on spurious correlations and generalize poorly to out - of - distribution (ood) data . recent work has explored using counterfactually - augmented data (cad)-data generated by minimally perturbing examples to flip the ground - truth label - to identify robust features that are invariant under distribution shift .

Seed term: diverse perturbation of examples

	unhelpful	helpful
Diverse perturbation of examples is used in order to generate counterfactual data that can help identify robust features that are invariant under distribution shift.	☐	☐
A counterfactual generator for diverse perturbation of examples.	☐	☐
We propose a method for generating CAD by diverse perturbation of examples.	☐	☐
In this paper, we propose a counterfactually - augmented data (cad) model that is robust to diverse perturbation of examples.	☐	☐
We use diverse perturbation of examples to flip the ground-truth label in order to identify robust features that are invariant under distribution shift.	☐	☐

Figure 7: Human evaluation example for GPT3.5Rnd, GPT3.5Retr, GPT3.5Rnd+CT, T5, and T5+SN+CL

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unhelpful

helpful

In that context, we use diverse perturbation of examples to generate counterfactually-augmented data, which helps identify robust features and improve the model's generalization to out-of-distribution data. This approach minimizes the model's reliance on spurious correlations and enhances its overall performance on natural language understanding tasks.



In that context, we use diverse perturbation of examples as a method because it helps in generating counterfactually-augmented data (CAD), which helps identify robust features that are invariant under distribution shift. By focusing on robust features, the model becomes less reliant on spurious correlations and can generalize better to out-of-distribution (OOD) data.



We propose a general framework based on diverse perturbation of examples that can be used to learn invariant features powerful enough to detect OOD data and reduce generalization error while performing natural language understanding tasks.



Figure 8: Human evaluation example for GPT3 . 5Rnd+KG, GPT4Rnd, and GPT4Rnd+KG